

# Misinformation Detection Using Transformer Language Models

P.3: Project Proposal  
AI and Natural Language Processing

Rahul Mishra  
MS Data Science, Northwestern University  
May 3, 2026

## Problem Statement and Motivation

This project addresses the critical challenge of automated multi-class veracity classification of political statements—distinguishing between six granular truth categories (*true*, *mostly-true*, *half-true*, *barely-true*, *false*, and *pants-on-fire*)—through fine-tuned transformer language models applied to the LIAR benchmark dataset. The fundamental problem combines three interrelated technical challenges: multi-class label imbalance across six ordinal veracity levels, short-text contextual representation where statements average fewer than twenty tokens, and the systematic integration of unstructured linguistic signals with structured speaker metadata to capture patterns of deception that transcend isolated claim analysis.

Automated misinformation detection represents both a pressing societal need and a technically demanding research frontier. Manual fact-checking pipelines—exemplified by organizations such as PolitiFact and FactCheck.org—cannot operate at the velocity and scale demanded by contemporary digital information ecosystems, where fabricated claims propagate across social networks within minutes and reach millions before corrections can be published. The asymmetry between the cost of producing disinformation and the cost of debunking it creates a structural vulnerability in democratic information environments. Natural language processing offers a pathway to scalable, real-time classification systems that can flag suspect content for human review, augment platform moderation queues, and provide probabilistic veracity signals to end users navigating contested informational terrain.

The six-class granularity of the LIAR dataset—capturing the spectrum from outright fabrication to literal truth—poses algorithmic challenges distinct from binary fake-news

classification. Prior benchmark results establish that traditional machine learning baselines (logistic regression, support vector machines, and LSTM architectures) plateau at approximately 27 percent test accuracy on the six-class task, barely exceeding the 16.7 percent random-choice floor. Recent advances in transformer-based language models, particularly BERT and its successors, demonstrate superior performance on nuanced classification tasks requiring fine-grained semantic discrimination, yet their application to political veracity classification with metadata fusion remains an active area of methodological development. This project contributes to that frontier by systematically evaluating whether contextual speaker features—party affiliation and cumulative credibility history—can overcome the accuracy ceiling that purely text-based classifiers encounter when decontextualized statements lack sufficient linguistic signals for confident categorization.

### Data Sources

Two publicly available benchmark datasets provide complementary training and evaluation resources for this project. The **LIAR dataset** (Wang 2017), available at [https://www.cs.ucsb.edu/~william/data/liar\\_dataset.zip](https://www.cs.ucsb.edu/~william/data/liar_dataset.zip), serves as the primary multi-class benchmark and contains 12,836 short political statements extracted from PolitiFact fact-checking articles, pre-partitioned into training (10,269 instances), validation (1,284 instances), and test (1,283 instances) splits. Each record comprises twelve structured fields: the natural-language statement text (averaging 17.9 tokens), a six-class veracity label, speaker name, party affiliation, speaker's home state, job title, a six-dimensional credit-history vector encoding the speaker's cumulative distribution of prior fact-check verdicts, subject category tags, and the context or venue in which the statement was originally made. This rich metadata structure enables experiments that fuse linguistic and speaker-level signals, directly testing whether biographical and historical features improve classification beyond statement text alone.

The **WELFake dataset** (Verma, Agrawal, and Patel 2021), distributed via Kaggle at <https://www.kaggle.com/datasets/saurabhshahane/fake-news-classification>, provides 72,134 full-length news articles—35,028 labeled authentic and 37,106 labeled fabricated—each consisting of a headline, article body text, and binary label. WELFake enables binary classification experiments and serves as a cross-dataset generalization benchmark, allowing evaluation of whether models trained on short political statements (LIAR) transfer to longer-form journalistic content, and conversely whether pre-training on high-volume binary data improves downstream multi-class performance.

Both datasets exhibit documented provenance, transparent labeling methodologies, and widespread adoption in the fake-news detection research community, ensuring reproducibility and facilitating direct performance comparisons with published baselines.

### **Preprocessing Methods**

**LIAR preprocessing** will normalize statement text through Unicode canonical decomposition (NFD normalization), case-folding to lowercase, and tokenization via the BERT WordPiece vocabulary with a maximum sequence length of 128 tokens—sufficient to capture 99.2 percent of statements without truncation based on corpus statistics. Class imbalance will be addressed through weighted cross-entropy loss, with per-class weights inversely proportional to empirical label frequencies in the training partition. The twelve structured metadata fields will be preserved in parallel feature vectors: party affiliation will be one-hot encoded across nine categories (Democrat, Republican, Independent, and six minor-party labels), and the six-dimensional credit-history vector will be L2-normalized to unit length and concatenated with party encodings to form a composite speaker representation. This dual-stream architecture—text processed through transformer layers, metadata through dense feedforward layers, both fused before the final classification head—enables controlled ablation studies isolating the marginal contribution of non-textual features.

**WELFake preprocessing** will concatenate article headlines and body text with a [SEP] separator token, tokenize using BERT WordPiece with a maximum sequence length of 512 tokens (truncating from the right to preserve lead paragraphs), and partition the full corpus into 80 percent training, 10 percent validation, and 10 percent test splits via stratified random sampling to preserve the original 51.4 percent fabricated-article base rate. Both datasets will be serialized as PyTorch Dataset objects with pre-computed input IDs, attention masks, and labels cached to disk, enabling GPU-accelerated batch loading and eliminating redundant tokenization overhead during iterative hyperparameter search.

All preprocessing transformations, random seeds, and data splits will be version-controlled and documented to ensure full experimental reproducibility.

### **Machine Learning Methods**

The core methodology comprises three progressive experimental phases, each building on the prior stage's findings.

**Phase 1: BERT Fine-Tuning Baseline.** The foundation model is bert-base-uncased (110 million parameters), initialized from Hugging Face pre-trained weights, with a trainable linear classification head projecting the 768-dimensional [CLS] token representation to six-class logits for LIAR or two-class logits for WELFake. Fine-tuning will employ the AdamW optimizer with learning rate  $2e-5$ , linear warmup over the first 10 percent of training steps, weight decay 0.01, and gradient clipping at norm 1.0. Training will proceed for up to 10 epochs with early stopping based on validation macro-F1 score monitored with patience of 3 epochs. This baseline establishes the performance ceiling achievable from linguistic features alone and provides calibrated predictions for subsequent fusion experiments.

A **TF-IDF + Logistic Regression baseline** will serve as the traditional machine-learning comparison point: statement text will be vectorized using term frequency-inverse document frequency weighting with 5,000 maximum features, bigram expansion, and L2 normalization, then classified via multinomial logistic regression with L2 regularization strength selected through five-fold cross-validation on the training set. This shallow baseline contextualizes the gains attributable to deep contextual representations.

**Phase 2: Metadata-Augmented Multi-Input Architecture.** Building on the BERT baseline, this phase introduces a dual-stream fusion architecture motivated by recent findings that graph-augmented and multi-modal models significantly outperform text-only classifiers (Moorthy et al. 2025). The text stream processes statement tokens through fine-tuned BERT layers, extracting the 768-dimensional [CLS] embedding. The metadata stream processes the concatenated speaker feature vector (one-hot party affiliation plus L2-normalized credit-history) through a two-layer feedforward network with 128 hidden units, ReLU activation, and dropout 0.3. Both streams are concatenated and passed through a final feedforward fusion layer (256 units, ReLU, dropout 0.3) before the six-class classification head. This architecture enables the model to learn non-linear interactions between linguistic deception patterns and speaker credibility profiles—for instance, whether vague quantitative claims are more indicative of falsehood when made by speakers with poor historical accuracy records.

**Phase 3: Hybrid Optimization and Advanced Transformer Variants.** The final experimental phase explores two complementary extensions. First, building on recent work demonstrating performance gains from hybrid optimization algorithms (Ganesh Karthik et al. 2025), the dual-stream model will be fine-tuned using an adaptive learning rate scheduler that combines gradient-based updates with population-based search over critical hyperparameters

(learning rate, dropout rate, fusion layer width). Second, the bert-base-uncased encoder will be replaced with RoBERTa-base and DeBERTa-v3-base to evaluate whether enhanced pre-training objectives and disentangled attention mechanisms improve veracity classification accuracy. Recent comparative studies indicate that DeBERTa variants excel on tasks requiring fine-grained semantic distinction (Alghamdi, Lin, and Luo 2025; Raza, Paulen-Patterson, and Ding 2025), making them well-suited to the six-class granularity of LIAR labels.

All experiments will be conducted on NVIDIA A100 GPUs with mixed-precision training (FP16) to accelerate convergence and reduce memory footprint.

### Evaluation Metrics

The primary evaluation metric is **macro-averaged F1-score**, computed as the unweighted mean of per-class F1 scores across the six veracity categories. Macro-F1 treats each class equally regardless of support, ensuring that model performance is not dominated by majority classes and that rare-but-critical categories such as *pants-on-fire* receive appropriate weight in the aggregate metric. This choice aligns with established practice in imbalanced multi-class benchmarks and facilitates direct comparison with published LIAR baseline results.

Secondary metrics include **test accuracy** (the proportion of correct predictions), **weighted F1-score** (to assess performance weighted by class frequency), **per-class precision and recall** (to diagnose category-specific failure modes), and **confusion matrices** (to identify systematic misclassification patterns such as conflation of adjacent veracity levels). For the binary WELFake task, **AUC-ROC** will supplement F1-score to measure discriminative power independent of classification threshold.

Cross-dataset generalization will be quantified by training on LIAR and evaluating on WELFake after label coarsening (collapsing the six LIAR classes to binary true/false), and vice versa, to assess whether learned representations transfer across statement length and domain.

### Related Work and Novelty

The LIAR dataset was introduced by Wang (2017), who demonstrated that traditional machine learning baselines achieve only modest accuracy (approximately 27 percent on six-class classification) and that incorporating speaker metadata provides complementary signal beyond statement text alone. The original work established the benchmark this project

targets for improvement but employed shallow feature-engineered models that cannot capture the deep contextual representations modern transformers afford.

Devlin et al. (2019) presented BERT, the bidirectional transformer pre-training paradigm whose contextual embeddings encode subtle linguistic patterns distinguishing truth from fabrication. BERT's masked language modeling objective enables it to capture long-range dependencies and nuanced semantics, making it markedly better suited to granular veracity classification than shallow bag-of-words approaches. However, the original BERT paper did not address fake-news detection or metadata fusion, leaving open the question of how to optimally integrate non-textual features.

Shu et al. (2017) systematically surveyed fake-news detection methods across linguistic, social-network, and user-level feature spaces, demonstrating that purely text-based classifiers plateau at moderate accuracy—a structural limitation motivating multi-modal approaches. Recent work has validated this insight: Moorthy et al. (2025) introduced a dual-stream graph-augmented transformer integrating BERT with graph neural networks to model information propagation structure, achieving 99 percent accuracy on social media fake-news benchmarks by fusing textual and relational features. Similarly, Ganesh Karthik et al. (2025) demonstrated that hybrid optimization algorithms—combining gradient descent with population-based search—significantly improve transformer fine-tuning for fake-news detection, achieving state-of-the-art results on multiple benchmarks.

The landscape of transformer-based misinformation detection has evolved rapidly in 2025. Alghamdi, Lin, and Luo (2025) proposed ABERT, a parameter-efficient adaptation of BERT using adapters that matches full fine-tuning performance with 67.7 percent fewer trainable parameters, demonstrating that efficiency gains need not compromise accuracy. Gondwe (2025) conducted a comprehensive evaluation of AI models for real-time misinformation detection, showing that hybrid approaches integrating machine learning, natural language processing, and deep learning outperform single-method baselines. Raza, Paulen-Patterson, and Ding (2025) performed a systematic comparison of BERT-like encoder models and large autoregressive language models (GPT-family) for fake-news classification, finding that encoder-only models generally outperform decoder-only models on classification tasks despite the latter's superior zero-shot generalization.

This project advances the state of the art by systematically combining three elements rarely integrated in prior work: (1) fine-tuning recent transformer variants (BERT, RoBERTa,

DeBERTa) on the LIAR multi-class benchmark with rigorous hyperparameter optimization, (2) fusing textual representations with structured speaker metadata through a trainable multi-input architecture, and (3) evaluating cross-dataset transfer between short-form political statements (LIAR) and long-form news articles (WELFake) to assess domain robustness. The dual-stream fusion architecture directly tests whether speaker credibility signals can overcome the accuracy ceiling that text-only models encounter when linguistic evidence is ambiguous—a hypothesis motivated by recent multi-modal successes but not yet validated on the LIAR benchmark with modern transformer encoders.

### **Expected Outcomes and Potential Challenges**

This project expects to achieve macro-averaged F1-scores exceeding 0.40 on the LIAR test set through BERT fine-tuning alone, representing a substantial improvement over the 0.27 accuracy baseline established by Wang (2017) using shallow models. The metadata-augmented dual-stream architecture is anticipated to yield an additional 3–5 percentage point gain by enabling the model to leverage speaker credibility patterns, with the largest improvements expected on ambiguous statements where linguistic signals alone provide insufficient evidence. On the binary WELFake benchmark, fine-tuned BERT models are expected to achieve F1-scores exceeding 0.93, consistent with recent reported results on similar datasets.

Several challenges may constrain performance. First, the LIAR dataset's six-class granularity creates inherent ambiguity: the boundary between *mostly-true* and *half-true*, for instance, reflects editorial judgment rather than objective ground truth, introducing label noise that limits the theoretical accuracy ceiling. Second, the credit-history metadata encodes cumulative speaker track records, which may introduce temporal confounding if speakers' truthfulness patterns evolve over time. Third, cross-dataset transfer experiments may reveal that models overfit to domain-specific lexical patterns (political jargon in LIAR, journalistic style in WELFake) rather than learning generalizable deception signals, necessitating domain adaptation techniques.

Addressing these challenges will require careful ablation studies isolating the contribution of each architectural component, error analysis identifying systematic failure modes through confusion matrices and per-class performance breakdowns, and potential incorporation of uncertainty quantification to flag low-confidence predictions for human review rather than forcing hard classifications on inherently ambiguous cases.

## References

- Alghamdi, J., Lin, Y., and Luo, S. 2025. ABERT: Adapting BERT Model for Efficient Detection of Human and AI-Generated Fake News. *International Journal of Information Management Data Insights* 15(1): 100353. <https://doi.org/10.1016/j.jjime.2025.100353>.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Volume 1, pp. 4171–4186. Minneapolis, MN: Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1423>.
- Ganesh Karthik, M., Khadri Syed Faizz Ahmad, Sai Geetha Pamidimukkala, Asha Prashant Sathe, Sirisha G N V G, Sitha Ram M, and Koteswararao Ch. 2025. Hybrid Optimization Driven Fake News Detection Using Reinforced Transformer Models. *Scientific Reports* 15(1): 14782. <https://doi.org/10.1038/s41598-025-99936-3>.
- Gondwe, G. 2025. Can AI Outsmart Fake News? Detecting Misinformation with AI Models in Real-Time. *Emerging Media* 3(2): 252–274. <https://doi.org/10.1177/27523543251325902>.
- Moorthy, H. R., Avinash, N. J., Krishnaraj Rao, N. S., Raghunandan, K. R., Dodmane, R., Blum, J. J., and Gabralla, L. A. 2025. Dual Stream Graph Augmented Transformer Model Integrating BERT and GNNs for Context Aware Fake News Detection. *Scientific Reports* 15(1): 25436. <https://doi.org/10.1038/s41598-025-05586-w>.
- Raza, S., Paulen-Patterson, D., and Ding, C. 2025. Fake News Detection: Comparative Evaluation of BERT-like Models and Large Language Models with Generative AI-Annotated Data. *Knowledge and Information Systems* 67(1): 1523–1551. <https://doi.org/10.1007/s10115-024-02321-1>.
- Shu, K., Sliva, A., Wang, S., Tang, J., and Liu, H. 2017. Fake News Detection on Social Media: A Data Mining Perspective. *ACM SIGKDD Explorations Newsletter* 19(1): 22–36. <https://doi.org/10.1145/3137597.3137600>.

Verma, P. K., Agrawal, P., and Patel, I. 2021. WELFake: Word Embedding over Linguistic Features for Fake News Detection. *IEEE Transactions on Computational Social Systems* 8(4): 881–893. <https://doi.org/10.1109/TCSS.2021.3068519>.

Wang, W. Y. 2017. "Liar, Liar Pants on Fire": A New Benchmark Dataset for Fake News Detection. *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, Volume 2: Short Papers, pp. 422–426. Vancouver, Canada: Association for Computational Linguistics. <https://doi.org/10.18653/v1/P17-2067>.